MITHIL K GOWDA

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Professional Summary

Cybersecurity graduate specializing in Digital Forensics, Incident Response, Threat Hunting, Malware Analysis, and Enterprise Security Operations. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on AI-driven cyber defense, intelligent threat detection, adaptive deception, and resilient security architectures. Passionate about forensic investigations, evidence preservation, incident analysis, and enterprise incident response. Seeking graduate opportunities in Digital Forensics, Incident Response, DFIR, and Enterprise Cyber Defense.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru 2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80 / 10**

Technical Skills

Digital Forensics: Disk Forensics, Memory Forensics, Network Forensics, Windows Forensics, Linux Forensics, Mobile Forensics Fundamentals, Evidence Collection, Chain of Custody, Timeline Analysis

Forensics Tools: Volatility, Autopsy, FTK Imager, KAPE, Redline, Sysinternals Suite, Velociraptor, CyberChef

Incident Response: Incident Detection, Incident Response Lifecycle, Threat Hunting, IOC Analysis, Malware Triage, Root Cause Analysis, Log Analysis, Live Response

Forensics Tools: Volatility, Autopsy, FTK Imager, KAPE, Redline, Wireshark, Sysinternals Suite

Detection Engineering: Sysmon, Windows Event Logs, Sigma Rules, YARA, MITRE ATT&CK, Cyber Kill Chain, IOC Development

Programming: Python, Java, JavaScript, SQL, Bash

Security Tools: Wireshark, Burp Suite, OWASP ZAP, Nmap

Networking: TCP/IP, DNS, HTTP/HTTPS, SSH, Network Traffic Analysis, PCAP Analysis, TCP Stream Analysis

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP)

Operating Systems: Windows, Linux

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL supporting secure engineering operations and enterprise analytics.
- Implemented authentication, authorization, secure API communication, and database security controls supporting enterprise software reliability.
- Developed scalable data processing pipelines supporting security analytics, operational investigations, structured log processing, and intelligent decision support.
- Engineered maintainable backend architectures emphasizing secure software engineering, operational resilience, forensic data integrity, and enterprise security.

Digital Forensics & Incident Response Projects

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good.*

- Designed an intelligent DFIR architecture integrating AI-driven threat detection, incident response automation, adaptive deception, threat hunting, and distributed threat intelligence.
- Developed machine learning models integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, reinforcement learning, IOC analysis, and intelligent incident response.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise security incident scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, 0.992 ROC-AUC, and a 1.85% false positive rate.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a secure digital platform integrating authentication, transaction integrity, trust scoring, auditability, evidence preservation, and fraud detection.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, secure authentication, transaction integrity verification, and cryptographic evidence protection.

C^3T-STAVP Cryptographic Framework (Ongoing Research)

- Designing an intelligent cybersecurity framework integrating Character Chaffing & Transposition Technology (C^3T), Secure Transaction Authorization & Verification Protocol (STAVP), adaptive trust validation, and resilient cyber defense.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, semantic obfuscation, forensic integrity, privacy-preserving architectures, and resilient incident response mechanisms.
- Developing intelligent cryptographic security mechanisms supporting digital evidence protection, distributed trust management, and enterprise forensic readiness.

Professional Certifications

- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- Google Cloud Computing – NPTEL
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL
- IBM Front-End Developer Specialization – IBM

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO - LEOS).
- First Author of internationally recognized research focused on AI-driven cybersecurity, intelligent distributed systems, autonomous cyber defense, and secure software architectures.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.